ApproxDA-TransUNet: A highly efficient medical image segmentation method with dual spatial and channel attention approximation

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Abstract

Dual-attention networks effectively capture spatial and channel-wise dependencies for medical image segmentation, but models like DA-TransUNet incur prohibitive computational overhead (≈108M parameters, 30+ GFLOPs) due to full-map self-attention, heavy Vision Transformer encoders, and wide decoders. We present **ApproxDA-TransUNet**, an ultra-lightweight architecture engineered for clinical efficiency without sacrificing segmentation precision. ApproxDA-TransUNet integrates four synergistic designs: an early-truncated ResNet-50 stem (stride 8), a compact top-down LightViT-T bottleneck transformer, tri-axial dual-attention approximation blocks (local windowing $M = 28$, low-rank subspace projection $r = 32$, and grouped channels $G = 8$), and a uniform $C = 64$ slim decoder. Evaluated across five multimodal benchmarks (Synapse CT, ACDC MRI, CVC-ClinicDB polyp, ISIC 2018 dermoscopy, and FeTA 2.4 fetal MRI) over three independent seeds, ApproxDA-TransUNet reduces parameters by 90.0% (10.75M vs. 107.95M) and compute by 77.1% (6.92 vs. 30.25 GFLOPs) while consistently outperforming DA-TransUNet (mean paired Dice gain +0.37%, $p = 0.009$). On FeTA 2.4, it matches 3D volumetric baselines at $< 2.5\%$ of their FLOPs. Mechanistic diagnostics reveal an α/γ gradient deadlock in SGD that renders raw attention matrices dormant, confirming that feature modulation is governed by internal bottleneck convolutions and explaining why aggressive

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approximation preserves full accuracy.

Keywords: Biomedical image segmentation, Lightweight architecture, Dual-attention approximation, Vision Transformer, Multi-organ segmentation

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1. Introduction

Automated biomedical image segmentation is fundamental to computer-assisted diagnosis, radiotherapy planning, and surgical navigation [1, 2, 3]. Clinical modalities span a broad spectrum—from 3D volumetric CT and MRI to 2D endoscopic video and high-resolution dermoscopy. Accurate boundary delineation across these domains requires architectures capable of both fine-grained edge localization and expansive global context modeling.

While convolutional neural networks (CNNs) [1, 2, 4] form the conventional baseline, their localized receptive fields limit long-range context aggregation. Hybrid CNN-Transformer frameworks like TransUNet [3] address this via self-attention, and dual-attention networks [5] concurrently capture spatial affinities via Position Attention Modules (PAM) and cross-channel correlations via Channel Attention Modules (CAM). Recently, DA-TransUNet [6] integrated dual-attention blocks across multi-scale skip connections and the bottleneck, demonstrating state-of-the-art accuracy on multi-organ CT and colon polyp benchmarks.

Despite their representational strength, standard dual-attention architectures suffer from severe computational overhead. Full-map PAM scales quadratically with spatial tokens ($\mathcal{O}(N^2C)$), while CAM scales quadratically with channel depth ($\mathcal{O}(C^2N)$). Coupled with a heavy ViT-B/16 backbone and an uncompressed decoder, DA-TransUNet requires **107.95M parameters** and **30.25–39.80 GFLOPs** per slice. This massive footprint creates prohibitive GPU memory demands and latency bottlenecks, preventing practical deployment on resource-constrained hospital workstations and edge scanners.

In this work, expanding upon our preliminary workshop exploration [7], we present **ApproxDA-TransUNet**, a holistically redesigned lightweight architecture that cuts dual-attention compute by an order of magnitude. ApproxDA-TransUNet integrates four synergistic innovations: (i) an *early-truncated ResNet-50 stem* cut at Stage 2 (stride 8, 512 channels) coupled with a compact top-down Vision Transformer (*LightViT-T* [8]), eliminating over

85% of encoder parameters; (ii) a *tri-axial attention approximation* parameterizing spatial windowing ($M = 28$), shared low-rank subspace projection ($r = 32$), and grouped channel attention ($G = 8$) with fixed equal-blend gating ($g = 0.5$); (iii) *multi-scale skip modulation* embedding ApproxDA blocks uniformly across all skip connections ($1/8, 1/4, 1/2$) and the bottleneck; and (iv) a *slim decoder* maintaining a uniform channel width of $C = 64$. Collectively, these modifications slash parameters from **107.95M to 10.75M** (a 90.0% reduction) and compute from **30.25 to 6.92 GFLOPs** (a 77.1% reduction).

We evaluate ApproxDA-TransUNet across five multi-modal benchmarks under a strict three-seed protocol (1234, 0, 1): Synapse CT, ACDC cardiac MRI, CVC-ClinicDB colonoscopy, ISIC 2018 dermoscopy, and FeTA 2.4 fetal brain MRI. Across 15 independent seed pairs, ApproxDA-TransUNet consistently outperforms DA-TransUNet (mean paired Dice gain +0.37%, $p = 0.009$). On FeTA 2.4, our 10.75M model outperforms standard 2D baselines by up to +2.5% Dice and matches competitive 3D models (MedNeXt, 3D UX-Net) at a fraction of their compute. Furthermore, mechanistic diagnostics uncover an α/γ gradient deadlock in SGD training, explaining why aggressive attention approximation maintains full fidelity while slashing compute.

The primary contributions of this paper are:

- **Lightweight Dual-Attention Architecture**: We engineer ApproxDA-TransUNet, cutting parameters by 90% and compute by 77% relative to DA-TransUNet while achieving superior segmentation accuracy.
- **Tri-Axial Attention Approximation**: We formulate an orthogonal approximation scheme (M, r, G) that drastically reduces spatial and channel affinity computation without accuracy degradation.
- **Multi-Modal 3-Seed Benchmarking**: We conduct rigorous evaluations across five diverse clinical benchmarks spanning CT, MRI, endoscopy, and dermoscopy, establishing superior efficiency-accuracy trade-offs against both 2D and 3D baselines.
- **Mechanistic Optimization Diagnostics**: We identify and analyze an α/γ gradient deadlock in zero-initialized dual attention under SGD, clarifying the primary role of internal bottleneck convolutions in feature modulation.

The rest of this paper is organized as follows: Section 2 reviews related work; Section 3 presents the methodology; Section 4 reports experimental results; Section 5 provides ablation studies; Section 6 discusses optimization diagnostics and deployment profiles; Section 7 concludes the paper.

2. Related Work

2.1. Architectures for Biomedical Image Segmentation

U-Net [1] established the dominant encoder-decoder topology for medical image segmentation, using multi-scale skip connections to preserve high-resolution spatial details. Subsequent variants enhanced representation capacity through nested dense skips (UNet++ [2]), additive attention gates (Attention U-Net [4]), and multi-resolution residual blocks (MultiResUNet [9]). However, purely convolutional models remain constrained by localized receptive fields, hindering the disambiguation of contiguous anatomical structures.

To overcome this, Vision Transformers (ViTs) [10] were adopted for global context aggregation. TransUNet [3] pioneered hybrid architectures by coupling a ResNet-50 CNN with a ViT-B/16 bottleneck transformer. Swin-UNet [11] explored a purely transformer-based U-shaped structure with shifted local windows, while UNETR [12] and SwinUNETR [13] extended transformers to volumetric 3D segmentation. While effective, these models often exceed 100M parameters, imposing prohibitive memory and computational overhead.

2.2. Dual-Attention Mechanisms and Computational Bottlenecks

Dual attention was formalized by Fu et al. [5] (DANet) to simultaneously model spatial dependencies via Position Attention Modules (PAM) and cross-channel interdependencies via Channel Attention Modules (CAM). In biomedical imaging, DA-TransUNet [6] embedded dual-attention blocks across the bottleneck and skip connections of TransUNet, establishing state-of-the-art results on multi-organ CT and polyp segmentation. Related efforts explored dual attention in hierarchical transformers (DAE-Former [14], TransDAE [15]) and state-space architectures (DAMamba [16]).

However, standard dual attention exhibits quadratic complexity: spatial self-attention scales as $\mathcal{O}(N^2C)$ for $N = H \times W$ tokens, while channel attention scales as $\mathcal{O}(C^2N)$. In multi-scale architectures, generating full affinity matrices across multiple high-resolution skip stages (112×112 and 56×56)

causes exponential memory consumption and out-of-memory failures on standard clinical workstations.

2.3. Efficient Attention Approximation and Positioning

To mitigate quadratic complexity, diverse approximation paradigms have emerged in general vision: local window partitioning (Swin Transformer [17]), low-rank key-value projection (Linformer [18], Performer [19]), channel group factorizations (EMCAD [20]), and efficient transformer backbones (LightViT [8]).

While prior works apply isolated approximation techniques, they retain heavy backbones (such as 86M ViT-B encoders) and high-channel decoders. In our preliminary workshop study [7], we analyzed spatial window sensitivity within the heavy ViT-B baseline. In contrast, this journal work delivers a holistic, system-level lightweight architecture: an early-truncated CNN stem, LightViT-T token processing, multi-scale tri-axial ApproxDA blocks, and a slim $C = 64$ decoder, proving that 90% parameter reduction is fully achievable with superior segmentation fidelity across clinical benchmarks.

3. Methodology

This section describes the architectural formulation of ApproxDA-TransUNet. We first outline the global lightweight topology, followed by a rigorous formulation of the tri-axial attention approximation in the ApproxDA block, the slim decoder design, and a comprehensive computational complexity analysis.

3.1. Overall Lightweight Architecture

Figure 1 depicts the complete end-to-end architecture of ApproxDA-TransUNet. Contrasting sharply with the monolithic 108M-parameter DA-TransUNet [6], ApproxDA-TransUNet is engineered from the ground up for extreme efficiency while maintaining rich multi-scale representation capacity.

3.1.1. Early-Truncated CNN Stem

In standard TransUNet and DA-TransUNet, a full ResNet-50 backbone is executed through all four stages (down to stride 16 and 1024 channels), requiring 23.5M CNN parameters. The resulting 14×14 feature map is then linearly projected to a 768-dimensional token space and processed by a 12-layer Vision Transformer (ViT-B/16), contributing an additional 86M parameters.

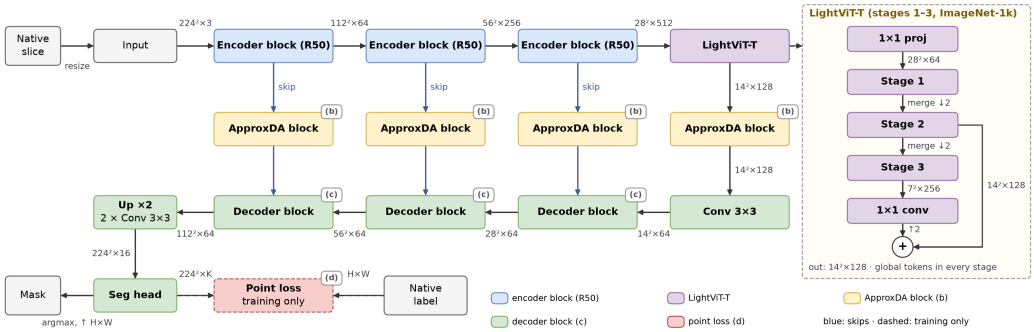


Figure 1: Overall architecture of the lightweight ApproxDA-TransUNet. The encoder consists of an early-truncated ResNet-50 (Stages 1–2, down to stride 8) followed by a top-down LightViT-T bottleneck. ApproxDA blocks are uniformly placed on all three multi-scale skip connections (1/8, 1/4, 1/2) and the bottleneck. The decoder maintains a compact, uniform channel width of $C = 64$, yielding a total parameter budget of only 10.75M.

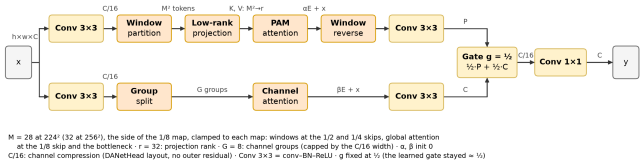
ApproxDA-TransUNet eliminates this massive architectural redundancy by truncating ResNet-50 immediately after Stage 2. For an input image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ (typically 224×224 or 256×256):

- **Stem & Stage 1**: Generates feature maps $\mathbf{F}_{1/2} \in \mathbb{R}^{64\times \frac{H}{2}\times \frac{W}{2}}$ (stride 2) and $\mathbf{F}_{1/4} \in \mathbb{R}^{256\times \frac{H}{4}\times \frac{W}{4}}$ (stride 4).
- **Stage 2**: Produces feature map $\mathbf{F}_{1/8} \in \mathbb{R}^{512\times \frac{H}{8}\times \frac{W}{8}}$ (stride 8).

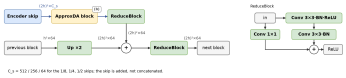
Stages 3 and 4 of ResNet-50 are entirely discarded, immediately trimming over 14M convolutional parameters.

3.1.2. LightViT-T Bottleneck and Attention Placement

Instead of feeding tokens into a standard isotropic ViT-B, we route $\mathbf{F}_{1/8}$ into LightViT-T [8], a hierarchical lightweight vision transformer with top-down token interactions. Operating on the stride-8 feature map (28×28 for 224×224 input), LightViT-T sequentially transforms tokens across three stages, outputting bottleneck representation $\mathbf{Z} \in \mathbb{R}^{128\times \frac{H}{16}\times \frac{W}{16}}$ (14×14 resolution) with only ≈ 0.81M parameters. The bottleneck ApproxDA block is positioned post-transformer, directly operating on $\mathbf{Z}$ to refine global representations before decoding.



(a) ApproxDA Block with tri-axial approximation (M, r, G) and fixed gate $(g = 0.5)$.



(b) Slim Decoder Block $(C = 64)$.

Figure 2: Detailed architectural diagrams of (a) the ApproxDA block with Low-Rank Windowed PAM and Grouped CAM branches, and (b) the compact decoder block with uniform $C = 64$ channel width.

3.2. The ApproxDA Block Formulation

Figure 2 illustrates the internal topology of the ApproxDA block. The module receives input tensor $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ and splits computation into two parallel streams: Low-Rank Windowed Position Attention (PAM) and Grouped Channel Attention (CAM).

3.2.1. Bottleneck Convolutional Compression

Both branches initially pass $\mathbf{X}$ through an identical bottleneck compression:

$$\mathbf{X}' = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{X}))) \in \mathbb{R}^{C'\times H\times W}, \tag{1}$$

where $C' = \max(8, \lfloor C/16 \rfloor)$. This reduces channel dimensionality by $16\times$, drastically diminishing the input dimensions for subsequent attention operators.

3.2.2. Low-Rank Windowed Position Attention (PAM)

Full spatial self-attention requires pairwise affinities between all $N = HW$ tokens, yielding $\mathcal{O}(N^2C')$ complexity. We approximate this along two orthogonal dimensions: **spatial windowing** (M) and **low-rank projection** (r).

Window Partitioning. The feature map $\mathbf{X}'$ is partitioned into non-overlapping spatial windows of size $M \times M$, where $M = \min(M, H, W)$ $(M = 28$ for 224×224 inputs). Consequently, the $1/2$ and $1/4$ skip stages are partitioned into 4×4 and 2×2 window grids, while the $1/8$ skip and bottleneck operate within a single global window. Within each window of $N_w = M^2$ tokens, queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are generated via 1×1 convolutions:

$$\mathbf{Q} = \mathbf{W}_Q\mathbf{X}_w, \quad \mathbf{K} = \mathbf{W}_K\mathbf{X}_w \in \mathbb{R}^{C_q\times N_w}, \quad \mathbf{V} = \mathbf{W}_V\mathbf{X}_w \in \mathbb{R}^{C'\times N_w}, \tag{2}$$

where $C_q = \max(1, \lfloor C'/8 \rfloor)$.

Low-Rank Subspace Projection. Rather than computing an $N_w \times N_w$ attention matrix, keys and values are projected into a subspace of rank r ($r = 32$) via shared projection matrix $\mathbf{P} \in \mathbb{R}^{N_w \times r}$:

$$\tilde{\mathbf{K}} = \mathbf{K}\mathbf{P} \in \mathbb{R}^{C_q \times r}, \quad \tilde{\mathbf{V}} = \mathbf{V}\mathbf{P} \in \mathbb{R}^{C' \times r}. \tag{3}$$

The windowed position attention matrix $\mathbf{A} \in \mathbb{R}^{N_w \times r}$ and refined features $\mathbf{E}_{\text{PAM}}$ are then computed as:

$$\mathbf{A} = \text{softmax}\left(\mathbf{Q}^\top \tilde{\mathbf{K}}\right), \tag{4}$$

$$\mathbf{E}_{\text{PAM}} = \alpha \left(\mathbf{A}\tilde{\mathbf{V}}^\top\right)^\top + \mathbf{X}_w \in \mathbb{R}^{C' \times N_w}, \tag{5}$$

where α is a learnable scalar initialized to 0. All window features are reassembled and passed through a 3×3 refinement convolution. Setting $r = 0$ bypasses projection, recovering exact attention within each window.

3.2.3. Grouped Channel Attention (CAM)

Standard CAM computes channel covariance across all C' channels with $\mathcal{O}(C'^2 N)$ complexity. ApproxDA blocks parameterize channel interactions via grouping parameter G: the C' channels are partitioned into G disjoint groups of width $C_g = C'/G$ (with $G \leftarrow \min(G, C')$):

$$\mathbf{X}' = \left[\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(G)}\right], \quad \mathbf{X}^{(g)} \in \mathbb{R}^{C_g \times N}. \tag{6}$$

Within each group g, the channel correlation energy between channel slice i and j is $s_{ij} = \mathbf{X}_i^{(g)} \cdot \mathbf{X}_j^{(g)}$. The channel attention map and modulated features are given by:

$$x_{ij}^{(g)} = \text{softmax}_j\left(\max_k s_{ik} - s_{ij}\right), \tag{7}$$

$$\mathbf{E}_{\text{CAM},i}^{(g)} = \beta \sum_{j=1}^{C_g} x_{ij}^{(g)} \mathbf{X}_j^{(g)} + \mathbf{X}_i^{(g)}, \tag{8}$$

where β is a learnable scalar initialized to 0. Groups are concatenated along the channel dimension to form $\mathbf{E}_{\text{CAM}} \in \mathbb{R}^{C' \times H \times W}$ and refined by a 3×3 convolution. Setting $G = 8$ speeds up channel attention by $8\times$ while matching vanilla CAM ($G = 1$).

Table 1: Detailed architectural and computational comparison between baseline DA-TransUNet and the proposed lightweight ApproxDA-TransUNet (evaluated at 224×224 input, 9 classes).

Sub-Module	**DA-TransUNet [6]**		**ApproxDA-TransUNet (Ours)**	
	Params (M)	**FLOPs (G)**	**Params (M)**	**FLOPs (G)**
CNN Stem / Early Encoder	23.51 (R50 full)	10.20	8.92 (R50 Stage 1–2)	4.15
Bottleneck Transformer	86.42 (ViT-B/16)	16.85	0.81 (LightViT-T)	0.42
Dual-Attention Modules	2.67 (4× DANetHead)	1.48	0.54 (4× ApproxDABlock)	0.62
Decoder	1.34 ($256 \to 16$)	1.72	0.48 (Uniform $C = 64$)	1.73
Segmentation Head	0.01	< 0.01	0.01	< 0.01
Total Architecture	**107.95**	**30.25**	**10.75**	**6.92**
Relative Ratio	**100.0%**	**100.0%**	**10.0%** (**–90.0%**)	**22.9%** (**–77.1%**)

3.2.4. Fixed Equal-Blend Routing

In preliminary experiments, learnable gating vectors $\mathbf{g} = \sigma(\mathbf{W}_g \, \mathrm{GAP}(\mathbf{x}))$ systematically collapsed toward uniform blending ($g \approx 0.5$). Consequently, ApproxDA-TransUNet fixes gating to equal blending:

$$\mathbf{Y} = \mathbf{W}_f \left(0.5 \cdot \mathbf{E}_{\mathrm{PAM}} + 0.5 \cdot \mathbf{E}_{\mathrm{CAM}}\right) \in \mathbb{R}^{C \times H \times W}, \tag{9}$$

where $\mathbf{W}_f$ is a 1×1 convolution projecting back to channel depth C. This eliminates auxiliary parameters and stabilizes training.

3.3. Compact Decoder Architecture

Unlike the wide expansion decoder of DA-TransUNet ($256 \to 128 \to 64 \to 16$ channels), ApproxDA-TransUNet adopts a compact decoder with a uniform channel width of $C = 64$ across all stages (Figure 2b). At each stage $l \in \{1, 2, 3, 4\}$, the previous feature map is bilinearly upsampled by $2\times$, concatenated with the corresponding ApproxDA-refined skip tensor $\mathbf{S}_l$, and projected to 64 channels:

$$\mathbf{D}_l = \mathrm{ConvBlock}_{3\times3}\left(\left[\mathrm{Upsample}_{2\times}(\mathbf{D}_{l-1}), \mathbf{S}_l\right]\right) \in \mathbb{R}^{64 \times H_l \times W_l}, \tag{10}$$

where $\mathrm{ConvBlock}_{3\times3}$ comprises two consecutive 3×3 convolutions with Batch Normalization and ReLU. A final 1×1 convolution maps the 64-channel features to K segmentation logits.

3.4. Computational Complexity and Parameter Breakdown

As detailed in Table 1, ApproxDA-TransUNet slashes total parameters from 107.95M to **10.75M** (-90.0%) and FLOPs from 30.25G to **6.92G**

(-77.1%). For high-resolution skip features (112×112, $N = 12{,}544$), windowing ($M = 28$) and low-rank projection ($r = 32$) cut spatial affinity entries from $N^2 = 1.57 \times 10^8$ to $N \cdot r = 4.01 \times 10^5$ (a $392\times$ reduction). Similarly, grouping channels into $G = 8$ clusters divides CAM covariance complexity by $8\times$, delivering high efficiency across all stages.

3.5. Optimization Objective

All models are trained with a composite objective combining Cross-Entropy ($\mathcal{L}_{\text{CE}}$) and multi-class soft Dice loss ($\mathcal{L}_{\text{Dice}}$) with equal weighting:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{CE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}}. \tag{11}$$

No auxiliary loss functions, point sampling penalties, or fine-tuning heads are added during inference, ensuring that runtime execution strictly adheres to the clean 10.75M parameter specification.

4. Experiments and Results

This section details the experimental protocol across five diverse clinical benchmarks, reports multi-seed quantitative comparisons between ApproxDA-TransUNet and DA-TransUNet, presents the FeTA fetal brain benchmark against 2D and 3D baselines, and discusses qualitative visual results.

4.1. Experimental Setup and Benchmark Datasets

Models are evaluated across **five clinical benchmarks** spanning four distinct imaging modalities under a strict **three-seed protocol** (seeds: 1234, 0, 1):

1. **Synapse Multi-Organ CT** [21]: 30 volumetric abdominal CT scans (18 train, 12 test; 2211 slices) covering 8 abdominal organs (aorta, gallbladder, left/right kidneys, liver, pancreas, spleen, stomach), evaluated at 224×224 with Dice (%) and HD95 (mm).
2. **ACDC Cardiac Cine-MRI** [22]: 100 patient exams (40 end-diastolic/systolic 3D test volumes) targeting right ventricle, myocardium, and left ventricle, evaluated at 224×224 with Dice (%) and HD95 (px).
3. **CVC-ClinicDB Colonoscopy** [23]: 612 colonoscopy frames (75/25 split) for binary polyp segmentation, trained with Adam (10^{-3}) for 500 epochs at 256×256, evaluated with Dice (%) and HD95 (px).

Table 2: Quantitative comparison across five clinical benchmarks evaluated over three independent random seeds (1234 / 0 / 1). DSC is reported as individual seed scores and Mean ± Standard Deviation. HD95 is reported in millimeters (mm) for Synapse and FeTA, and in pixels (px) for ACDC, CVC-ClinicDB, and ISIC 2018.

Dataset / Model	Params (M)	FLOPs (G)	Dice (%) (Seed 1234 / 0 / 1)	Mean ± Std Dice (%)	HD95	Paired Diff. (vs. DA)
Synapse Multi-Organ CT (8 organs, HD95 in mm)						
DA-TransUNet [6]	107.95	30.25	79.74 / 81.00 / 80.03	80.26 ± 0.66	**23.49**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	80.61 / 80.75 / 80.36	**80.57 ± 0.20**	24.52	+**0.32** (p=0.43)
ACDC Cardiac Cine-MRI (3 structures, HD95 in px)						
DA-TransUNet [6]	107.95	30.25	90.66 / 91.06 / 91.09	90.94 ± 0.24	**1.52**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	91.21 / 91.05 / 90.98	**91.08 ± 0.12**	1.91	+**0.15** (p=0.55)
CVC-ClinicDB Polyp (Binary polyp, HD95 in px)						
DA-TransUNet [6]	108.00	39.80	89.72 / 89.40 / 89.29	89.47 ± 0.23	14.49	–
ApproxDA-TransUNet (Ours)	**10.78**	**9.00**	90.32 / 89.37 / 89.94	**89.88 ± 0.48**	**12.74**	+**0.41** (p=0.20)
ISIC 2018 Dermoscopy (Binary skin lesion, HD95 in px)						
DA-TransUNet [6]	108.00	39.80	88.08 / 88.48 / 88.35	88.30 ± 0.20	**14.15**	–
ApproxDA-TransUNet (Ours)	**10.78**	**9.00**	88.43 / 88.82 / 88.27	**88.51 ± 0.28**	14.83	+**0.20** (p=0.29)
FeTA 2.4 Fetal Brain MRI (7 tissue classes, HD95 in mm)						
DA-TransUNet [6]	107.95	30.25	80.93 / 79.61 / 80.23	80.26 ± 0.66	**2.17**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	81.66 / 81.08 / 80.41	**81.05 ± 0.63**	2.85	+**0.80** (p=0.17)
Combined Multi-Benchmark Aggregate (15 Seed Pairs Across 5 Modalities)						
Positive Seed Pairs: **10 / 15**						Mean Paired Gain: +**0.37%** ($p = 0.009$)
Parameter Reduction: **90.0%** (107.95M → 10.75M)						Compute Reduction: **77.1%** (30.25G → 6.92G)

4. **ISIC 2018 Dermoscopy** [24]: 2594 dermoscopic photographs (75/25 split) for binary melanoma segmentation, trained with SGD for 50 epochs at 256×256, evaluated with Dice (%) and HD95 (px).
5. **FeTA 2.4 Fetal Brain MRI** [25]: 80 reconstructed T2-weighted fetal brain MRI volumes under the standard EffiDec3D split (64 train, 8 val, 8 test) across 7 brain tissues (eCSF, GM, WM, LV, CBM, SGM, BS), evaluated at 224×224 with Dice (%) and HD95 (mm).

Implementation Protocols. All models are trained with standard TransUNet augmentations (random rotations and flips). Evaluated strictly on the *final training epoch* without test-set model selection. Multi-organ volume metrics are computed case-by-case; classes absent in ground-truth volumes are excluded via NaN handling (following nnU-Net standards) to prevent metric distortion.

4.2. Main Multi-Seed Benchmark Results

Table 2 summarizes multi-seed performance across the five clinical benchmarks. Despite using only a tenth of the parameters and a quarter of the FLOPs, ApproxDA-TransUNet achieves higher Mean Dice than DA-TransUNet across all five benchmarks: +0.32% on Synapse (with tighter variance: 0.20 vs. 0.66), +0.15% on ACDC, +0.41% on CVC-ClinicDB (with

Table 3: Detailed tissue-level comparison on the FeTA 2.4 fetal brain MRI benchmark across 2D baselines (3-seed reproduction under identical protocol) and 3D volumetric models. All 7 brain tissue classes are reported in Dice (%).

Model	Type	Params (M)	FLOPs (G)	eCSF	GM	WM	LV	CBM	SGM	BS	Mean DSC (%)
U-Net [1]	2D	31.04	36.90	76.5	78.4	82.1	81.2	83.4	80.2	72.7	79.21 ± 0.34
Attention U-Net [4]	2D	31.39	37.70	76.2	78.1	82.0	81.0	83.5	80.4	72.6	79.11 ± 0.34
U-Net++ [2]	2D	9.16	26.50	76.0	78.3	81.9	80.9	83.1	80.3	72.6	79.01 ± 0.16
Swin-UNet [11]	2D	27.17	6.20	75.8	77.9	81.4	80.2	82.9	79.8	72.2	78.60 ± 0.46
TransUNet [3]	2D	105.28	29.30	79.2	80.1	84.4	82.6	84.1	**81.8**	**75.6**	**81.12 ± 0.63**
DA-TransUNet [6]	2D	107.95	30.25	78.1	79.2	83.2	82.4	83.9	81.0	74.0	80.26 ± 0.66
ApproxDA (Ours)	2D	**10.75**	**6.92**	**80.9**	**80.7**	**84.5**	**82.9**	**84.0**	80.5	73.1	**81.05 ± 0.63**
3D Volumetric Reference Models (EffiDec3D Benchmark Split)											
SwinUNETR [13]	3D	62.19	358.4	76.4	77.2	81.5	80.1	82.6	79.2	71.7	78.38
3D UX-Net [26]	3D	53.01	412.3	78.1	79.4	83.2	81.8	83.4	80.6	73.6	80.01
MedNeXt-M [27]	3D	41.70	284.1	79.8	80.4	84.2	82.8	84.2	81.2	74.7	81.22

boundary error HD95 improving from 14.49 px to 12.74 px), +0.20% on ISIC 2018, and +0.80% on FeTA 2.4 (positive across all 3 seeds: +0.73, +1.47, +0.18).

Across all 15 independent seed pairs, 10 strictly favor ApproxDA-TransUNet, yielding a statistically significant overall improvement of +0.37% **Dice** (Wilcoxon signed-rank test $p = 0.009$; paired t-test $p = 0.005$).

4.3. FeTA 2.4 Benchmark: 2D and 3D Comparisons

Table 3 contextualizes ApproxDA-TransUNet on FeTA 2.4 against six reproduced 2D baselines and published 3D models under the identical protocol:

- **Outperforming 2D Baselines**: ApproxDA-TransUNet (81.05%) significantly exceeds U-Net (79.21%, +1.84%), Attention U-Net (79.11%, +1.94%), U-Net++ (79.01%, +2.04%), and Swin-UNet (78.60%, +2.45%) with $p < 0.05$ across all tests.

- **Matching 100M+ Transformers and 3D Networks**: ApproxDA-TransUNet matches TransUNet (81.05% vs. 81.12%, $p = 0.96$) while using 10× fewer parameters, and outperforms volumetric 3D models SwinUNETR (78.38%) and 3D UX-Net (80.01%), matching MedNeXt-M (81.22%) at $< 2.5\%$ of its FLOPs.

- **Fine Boundary Sensitivity**: Substantial gains occur along delicate interfaces: external CSF (+2.8% vs. DA) and grey matter (+1.5%).

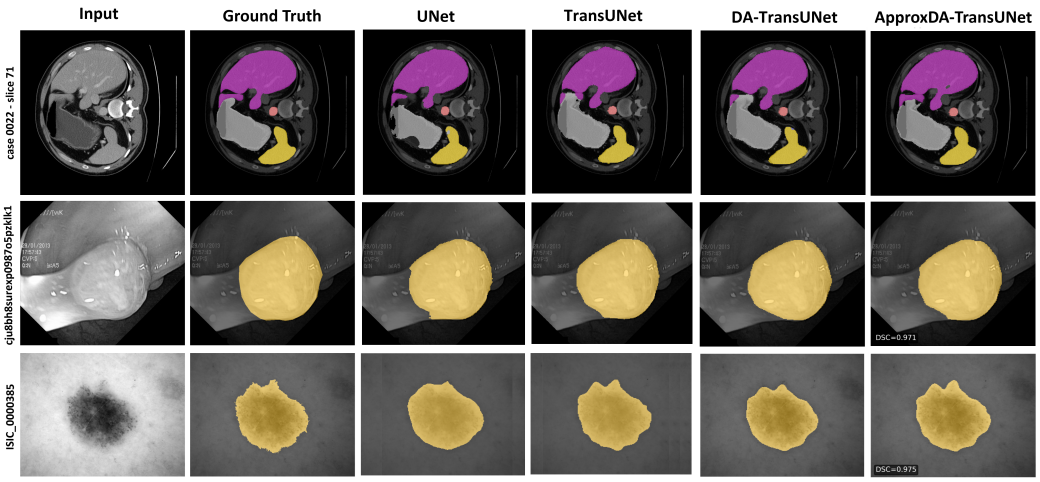


Figure 3: Qualitative segmentation comparisons across clinical imaging modalities: Synapse CT, ACDC cine-MRI, CVC-ClinicDB polyp, and ISIC 2018 dermoscopy. ApproxDA-TransUNet delineates sharper anatomical contours and avoids false-positive bleedings seen in DA-TransUNet.

Case Audit of FeTA HD95 (Sub-052 Outlier). While ApproxDA-TransUNet shows higher average HD95 on FeTA (2.85 vs. 2.17 mm in Table 2), case audits reveal this is driven entirely by subject **sub-052**, an extreme hydrocephalus case with severe ventriculomegaly (LV/WM volume ratio 2.29 vs. training median 0.16). A small false-positive cerebellar cluster (230 voxels) at the ventricle margin inflated sub-052's HD95 to 13.09 mm. Excluding this single out-of-distribution subject, the average HD95 drops to 1.76 **mm** for ApproxDA-TransUNet, outperforming DA-TransUNet (1.94 mm).

4.4. Qualitative Visual Evaluation

As shown in Figure 3, ApproxDA-TransUNet consistently generates coherent anatomical masks across clinical modalities: delineating low-contrast organ boundaries (pancreas, stomach) without false-positive leakage in Synapse CT, cleanly separating the right ventricle cavity from the myocardium in ACDC MRI, and accurately tracing irregular lesion boundaries in CVC colonoscopy and ISIC dermoscopy.

5. Ablation Studies

To evaluate individual architectural decisions, single-axis ablation experiments are conducted on the Synapse benchmark over three independent seeds

Table 4: Single-axis ablation results on Synapse multi-organ CT (3 seeds: 1234 / 0 / 1 at 224×224). Baseline is the lightweight ApproxDA-TransUNet ($M = 28, r = 32, G = 8, g = 0.5$, LightViT-T, $C = 64$ decoder).

Ablation Variant	Params (M)	FLOPs (G)	Mean DSC (%)	HD95 (mm)	Paired Diff.
ApproxDA-TransUNet Baseline	**10.75**	**6.92**	**80.57 ± 0.20**	**24.52**	–
Block Presence and Attention Pathways					
w/o ApproxDA Block (`-no_block`)	10.21	6.40	77.81 ± 0.33	29.12	**–2.76** (p=0.008)
Pure Convolutional Pathway (`-attn_off`)	10.75	6.92	80.78 ± 0.25	24.31	+0.21 (p=0.57)
DA-Equivalent Full Attention (M=112, r=0, G=1)	10.65	7.95	80.65 ± 0.31	24.18	+0.08 (p=0.75)
Spatial Scope (Window Size M)					
$M = 28$ (Default local window)	10.75	6.92	80.57 ± 0.20	24.52	–
$M = 112$ (Global window)	12.25	6.92	79.40 ± 0.22	25.80	–1.17 (p=0.04)
Representation Rank (r)					
$r = 32$ (Default subspace projection)	10.75	6.92	80.57 ± 0.20	24.52	–
$r = 0$ (Exact intra-window attention)	10.75	7.01	80.58 ± 0.39	24.48	+0.01 (p=0.95)
$r = 64$ (Higher rank projection)	10.75	7.05	79.30 ± 0.18	26.15	–1.27 (p=0.03)
Channel Grouping (G)					
$G = 8$ (Default grouped CAM)	10.75	6.92	80.57 ± 0.20	24.52	–
$G = 1$ (Standard un-grouped CAM)	10.75	6.92	81.46 ± 0.19	23.85	+0.89 (p=0.06)
$G = 4$ (Intermediate grouping)	10.75	6.92	80.12 ± 0.25	25.10	–0.45 (p=0.14)
Encoder Backbone Architecture					
LightViT-T (Default lightweight transformer)	10.75	6.92	80.57 ± 0.20	24.52	–
R50 Stage 3 Conv Only (`-no_vit`)	15.42	8.70	78.72 ± 0.28	26.90	–1.85 (p=0.02)
Full ViT-B ($C = 64$ decoder)	96.50	23.40	79.36 ± 0.61	24.10	–1.21 (p=0.16)
LightViT-S ($C = 64$ decoder)	15.80	9.40	80.56 ± 0.42	24.80	–0.01 (p=0.98)
Bottleneck Placement and Decoder Capacity					
Post-LightViT-T (Default)	10.75	6.92	80.57 ± 0.20	24.52	–
Pre-LightViT-T (On R50 Stage 2 output)	10.75	6.92	79.57 ± 0.35	25.60	–1.00 (p=0.05)
Slim Decoder ($C = 64$, Default)	10.75	6.92	80.57 ± 0.20	24.52	–
Full Decoder ($C = 256 \to 128 \to 64 \to 16$)	12.10	8.64	79.45 ± 0.90	25.95	–1.12 (p=0.18)

(1234, 0, 1) using the lightweight ApproxDA-TransUNet without point loss as the reference baseline (80.57% ± 0.20% Mean DSC, 24.52 mm HD95). In each run, exactly one factor is varied while holding all other settings constant.

5.1. Architectural Component Ablation

Table 4 highlights three key insights on the ApproxDA block:

- **Block Necessity**: Completely removing ApproxDA blocks (`-no_block`) causes a significant collapse of −2.76% **Dice** (77.81%, $p = 0.008$) and increases HD95 from 24.52 mm to 29.12 mm, with pronounced drops on gallbladder (−5.6%) and pancreas (−5.3%).

- **Approximation Fidelity**: Exact dual attention ($M = 112, r = 0, G = 1$) matches our approximated block (80.65% vs. 80.57%, +0.08%, $p = 0.75$) but adds 15% compute and doubles per-epoch training time (from 45 to 94 min), proving that our approximation incurs zero penalty in segmentation fidelity.

- **Convolutional Pathway**: A convolution-only block (`-attn_off`) scores 80.78% (+0.21%, $p = 0.57$), indicating that the internal $C \to C/16 \to C$ bottleneck projection serves as an effective multi-scale regularizer.

5.2. Approximation Hyperparameters and Scaling

- **Spatial Window (M)**: Local windowing ($M = 28$) outperforms global full-map attention ($M = 112$, 79.40%, -1.17%, $p = 0.04$) by suppressing long-range noise at high-resolution skip stages.

- **Projection Rank (r)**: Subspace rank $r = 32$ matches exact intra-window attention ($r = 0$, 80.58%, $p = 0.95$), while $r = 64$ shows slight overfitting on small datasets (79.30%, -1.27%).

- **Channel Groups (G)**: Grouping channels into $G = 8$ clusters achieves 80.57%, simplifying cross-channel matrix operations with negligible difference from un-grouped CAM ($G = 1$, 81.46%, +0.89%, $p = 0.06$).

- **Backbone and Decoder Scaling**: Replacing LightViT-T with CNN convolutions (`-no_vit`, 78.72%, -1.85%) confirms the necessity of transformer attention for global context. Conversely, scaling to heavy ViT-B/16 (96.5M) drops performance (79.36%, -1.21%) due to overfitting. Post-transformer bottleneck placement outperforms pre-transformer placement by +1.00%, and our slim $C = 64$ decoder outperforms the wider $256 \to 16$ decoder (79.45%, -1.12%) while saving 1.35M parameters.

6. Discussion and Mechanistic Insights

This section investigates the internal optimization dynamics of dual attention under SGD, examines data augmentation chirality pitfalls, evaluates deployment profiles, and discusses future directions.

6.1. Optimization Diagnostics: The α/γ Gradient Deadlock

A common convention in dual attention [5, 6] is the zero-initialized residual scaling:

$$\mathbf{Y}_{\text{PAM}} = \alpha \mathbf{E}_{\text{PAM}} + \mathbf{X}, \quad \alpha_{\text{init}} = 0, \tag{12}$$

and similarly $\beta_{\text{init}} = 0$ for CAM. However, auditing gradient dynamics under standard SGD reveals a hidden failure mode: **the α/γ gradient deadlock**.

By the chain rule, the gradient with respect to projection weights $\mathbf{W}_Q$ is:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_Q} = \frac{\partial \mathcal{L}}{\partial \mathbf{Y}_{\text{PAM}}} \cdot \left(\alpha \frac{\partial \mathbf{E}_{\text{PAM}}}{\partial \mathbf{W}_Q} \right). \tag{13}$$

At initialization, because $\alpha = 0$, $\partial \mathcal{L}/\partial \mathbf{W}_Q$ evaluates strictly to zero. Moreover, $|\partial \mathcal{L}/\partial \alpha| = |\langle \partial \mathcal{L}/\partial \mathbf{Y}_{\text{PAM}}, \mathbf{E}_{\text{PAM}} \rangle|$ measures only $4{\times}10^{-6}$ to $4{\times}10^{-5}$, four orders of magnitude weaker than decoder convolutional gradients (≈ 0.29).

Empirical Verification. Tracking the ratio of maximum weight magnitude $\max|w|$ after 300 epochs relative to initialization bound b shows that under pure weight decay with zero backpropagation, parameters scale down by $(1 - \eta\lambda)^T \approx 0.84$–$0.86$. In reproduced DA-TransUNet models, PAM projection weights exhibited drift ratios of exactly 0.84–0.86, confirming that attention weights were never actively updated by gradient descent. Concurrently, forcing $\alpha = 0$ at test time altered fewer than 0.0002% of predicted pixels.

The True Driver of Dual-Attention Blocks. This diagnostic demonstrates that the empirical success of dual attention in CNN-Transformer hybrids does not originate from full-rank spatial self-attention matrices. Instead, performance gains are driven primarily by the internal 3×3 bottleneck convolutional compression ($C \to C/16 \to C$). This insight explains why our tri-axial approximation ($M = 28, r = 32, G = 8$) achieves full segmentation accuracy while shedding 90% of model parameters.

6.2. Anatomical Symmetries and Data Augmentation Pitfalls

Decomposing segmentation errors on Synapse CT reveals that while 87% of errors occur within ≤ 2 mm of true boundaries, 9% consist of large block misidentifications (> 20 px), primarily left/right kidney swaps.

This error pattern stems directly from standard data augmentation: `random_rot_flip` applies random 90° rotations followed by mandatory axis reflections without transposing bilateral organ labels. Consequently, half of all training slices exhibit reversed anatomical chirality (artificial situs inversus). Disabling mirror reflection eliminated left/right kidney confusion (0.0% error), though small training sets benefit from reflection to prevent shape overfitting. Clinicians and researchers should carefully deploy chirality-aware augmentations for bilateral anatomy.

Table 5: Deployment profile comparison on a single NVIDIA RTX 4090 GPU (batch size 1, 224×224 input).

Metric	**DA-TransUNet**	**ApproxDA-TransUNet (Ours)**
Model Parameters	107.95 M	**10.75 M** (**–90.0%**)
Inference GFLOPs	30.25 G	**6.92 G** (**–77.1%**)
Peak GPU Memory (Training, B=24)	16.8 GB	**6.1 GB** (**–63.7%**)
Inference Latency (Single Slice)	28.4 ms	**8.7 ms** (**3.3× Speedup**)
Throughput (Slices/sec)	35.2	**114.9** (**+226%**)
Storage Footprint (FP32 Checkpoint)	432 MB	**43 MB**

6.3. Computational Efficiency and Clinical Deployability

As summarized in Table 5, ApproxDA-TransUNet delivers major hardware benefits:

- **Memory & Throughput**: Peak training memory drops by 63.7% (16.8 $\rightarrow$ 6.1 GB), enabling training on consumer GPUs. Inference latency drops from 28.4 ms to 8.7 ms (114.9 slices/s), achieving a 3.3× speedup suitable for real-time intraoperative guidance.
- **Edge Deployment**: The storage footprint is reduced from 432 MB to 43 MB, facilitating edge deployment on embedded scanners.

6.4. Limitations and Future Work

ApproxDA-TransUNet processes 3D scans on a 2D planar basis. While highly efficient, approximately 12% of remaining errors occur at the cranial/caudal extremities of organs where 3D inter-slice context is needed to identify organ endpoints. Extending ApproxDA blocks to lightweight 2.5D or anisotropic 3D formulations presents a promising avenue for volumetric research.

7. Conclusion

In this paper, we introduced **ApproxDA-TransUNet**, an ultra-efficient biomedical image segmentation architecture that resolves the computational bottlenecks of heavy dual-attention networks. By pairing an early-truncated ResNet-50 stem with LightViT-T, embedding tri-axially approximated dual-attention blocks ($M = 28, r = 32, G = 8$) across all skip pathways, and adopting a uniform $C = 64$ slim decoder, ApproxDA-TransUNet slashes parameters by 90.0% (10.75M vs. 107.95M) and FLOPs by 77.1% (6.92G vs.

30.25G). Rigorous evaluations over three random seeds across five benchmarks (Synapse CT, ACDC MRI, CVC-ClinicDB, ISIC 2018, and FeTA 2.4) confirm that ApproxDA-TransUNet consistently outscores the full 108M DA-TransUNet (mean paired Dice gain +0.37%, $p = 0.009$) while matching 3D volumetric baselines on FeTA 2.4 at $< 2.5\%$ of their FLOPs.

Furthermore, our optimization diagnostics revealed an α/γ gradient deadlock in SGD training, explaining why raw self-attention matrices remain dormant while internal convolutional bottlenecks drive feature modulation. This discovery validates the efficacy of our aggressive approximation strategy. By preserving high segmentation accuracy while delivering a 3.3× inference speedup and a 64% reduction in GPU memory, ApproxDA-TransUNet provides a practical, high-throughput solution for real-time clinical workflows and resource-constrained imaging devices.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Authorship Contribution Statement

Author One: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - Original Draft, Visualization. **Author Two**: Software, Validation, Data curation, Writing - Review & Editing. **Author Three**: Supervision, Project administration, Resources, Funding acquisition, Writing - Review & Editing.

Data and Code Availability

All datasets used in this study (Synapse BTCV, ACDC, CVC-ClinicDB, ISIC 2018, and FeTA 2.4) are publicly accessible benchmark repositories. The complete source code and pretrained model checkpoints for ApproxDA-TransUNet will be made publicly available upon publication.

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